MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCCSC202 - PYTHON FOR DATA ANALYTICS
QP CODE 25900324
(2024 ADMISSION ONWARDS) - **SCHEME FOR VALUATION**

Part A Very Short Answer Questions Answer all questions: Each question carries **2** marks			
1	**Among descriptive, diagnostic, predictive, and prescriptive analysis, which type focuses on "future actions to take"? Justify.**	[U] / [CO1]	1 mark (Name) + 1 mark (Justification)
Ans:	Prescriptive analysis. Because it goes beyond prediction to recommend the optimal course of action.		
2	**Define qualitative data. Give an example.**	[K] / [CO1]	1 mark (Definition) + 1 mark (Example)
Ans:	Definition: Data that describes characteristics or qualities and cannot be measured numerically. Example: Color, Gender.		
3	**Write the Python code to sort a NumPy array named arr.**	[K] / [CO2]	1 mark (Function usage) + 1 mark (Correct Syntax/Context)
Ans:	import numpy as np arr = np.array([1,2,3]) sorted_arr = np.sort(arr)		
4	**Name the pyplot function used to clear the current figure. Write an example.**	[K] / [CO3]	1 mark (Function Name) + 1 mark (Example)
Ans:	Function: plt.clf() Example: import matplotlib.pyplot as plt plt.clf()		
5	**Define a Pandas Series.**	[K] / [CO3]	2 marks (Comprehensive definition)
Ans:	A one-dimensional labeled array capable of holding data of any type, with labels (index).		
Part B Short Answer Questions Answer any **5** questions: Each question carries **5** marks			
6	**Discuss the importance of defining objectives in data analysis.**	[U] / [CO1]	1 mark for stating the importance+1 mark per strong point (Min 4 points needed)
Ans	Key Points: - Direction and focus - Scope definition - Success metrics - Stakeholder alignment		
7	**Explain array indexing in NumPy. Illustrate your answer with suitable examples.**	[U] / [CO2]	2 marks (Conceptual explanation) + 3 marks (Examples and illustration)
Ans:	Indexing types: basic, slicing, multidimensional, boolean Examples:		

	import numpy as np arr = np.array([1,2,3,4]) print(arr[1]) arr2 = np.array([[1,2],[3,4]]) print(arr2[1,0])		
8	**Explain the importance of vectorization in NumPy with an example.**	[U] / [CO2]	2 marks (Definition/Mechanism) + 1 mark (Importance points) + 2 marks (Clear example)
Ans:	Vectorization applies operations on whole arrays without explicit Python loops. Importance: faster, cleaner code, better memory locality. Example: import numpy as np arr = np.array([1,2,3]) print(arr+5)		
9	**Discuss the use of np.min(), np.max(), np.argmin(), and np.argmax() in analyzing datasets. Provide Python examples.**	[U] / CO2]	1 mark (Definition of functions) + 1 mark (Discussion/Use cases) + 3 marks (Four correct code examples)
Ans	**np.min()** – Returns the **smallest value** in the dataset. **np.max()** – Returns the **largest value** in the dataset. **np.argmin()** – Returns the **index/position of the smallest value**. **np.argmax()** – Returns the **index/position of the largest value**.		
10	**Explain the role of the pyplot module in matplotlib with an example**	[U] / CO3]	2 marks (Role/Functionality) + 3 marks (Comprehensive example demonstrating `plt.plot` and `plt.show`)
Ans:	Pyplot provides a MATLAB-like interface for plotting. Example: import matplotlib.pyplot as plt x=[1,2,3]; y=[2,4,1] plt.plot(x,y) plt.show()		
11	**Write the Python code to compute correlation between math, science, and english scores.** **data = {'math': [80, 90, 85], 'science': [75, 88, 82], 'english': [78, 85, 80]}**	[A] /[CO3]	1 mark (Import/Data setup) + 2 marks (DataFrame creation) + 2 marks (Correct correlation calculation)
Ans:	import pandas as pd data = {'math': [80, 90, 85], 'science': [75, 88, 82], 'english': [78, 85, 80]} df = pd.DataFrame(data) df.corr()		
12	**Write a Python code to detect and count missing values in a DataFrame.**	[A] / [CO3]	2 marks (Detection - `isnull()`) + 3 marks (Counting across
Ans	import pandas as pd		

	df=pd.DataFrame({'a':[1,None,3],'b':[4,5,None]}) print(df.isnull()) print(df.isnull().sum()) (Detection can be using `df.isnull()` or `df.isna()`)		columns - `.sum()` and context setup)
Part C Essay Questions Answer any **1** questions: Each question carries **15** marks			
13	**Explain the importance of dtype in NumPy arrays. How does dtype affect memory usage, computation speed, and type conversion (upcasting and downcasting) during array operations?**	[U] / [CO2]	Total 15 Marks based on depth and examples provided.
Ans:	Discuss memory usage, speed, upcasting, and downcasting. Include examples with int32, float64.		
14	**Explain single element selection, multiple element selection, and slicing from a Pandas Series and DataFrame. Provide suitable examples.**	[U] / [CO3]	Series – 5 marks (single: 2, multiple: 1, slicing: 2) DataFrame – 10 marks (single col: 2, multiple col: 2, .loc: 3, .iloc: 3)
Ans:	Series: single element selection using label/index Series: multiple element selection using list of labels/positions Series slicing using label-based or positional range DataFrame: single column selection DataFrame: multiple column selection DataFrame row/column access and slicing using .loc (label-based) DataFrame row/column access and slicing using .iloc (index-based)`		